INCH-POUND

MIL-PRF-1/1336H
w/AMENDMENT 1
22 March 2017
SUPERSEDING
MIL-PRF-1/1336H
16 September 2016

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, PULSE MODULATOR
# TYPE 3D21WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Beam tetrode, vacuum
Outline: See figure 1.
Cathode: Coated unipotential.
Base connections:

| Pin No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Cap |
|---|---|---|---|---|---|---|---|---|---|
| Element | hct | h | nc | g2 | nc | g1 | h | k | a |

ABSOLUTE RATINGS:

| Parameter: | Ef | Eb | Ec1 | Ec2 | Ehk | Pg1 | Ik | Ic1 |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | V dc | v | W | mA dc | mA dc |
| Maximum:<br>Minimum: | 13.9 or 6.9<br>11.4 or 5.7 | 4,000 1/<br>--- | 0<br>-200 2/ | 850<br>--- 3/ | 500<br>--- | 0.5<br>--- | 175<br>--- | 5.0<br>--- |
| Test condition: | 6.3 | 600 | 0 | 300 | --- | --- | --- | --- |

ABSOLUTE RATINGS:

| Parameter: | Pp | Pg2 | egk | ik | tk | TE | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | W | W | v | a | sec | °C | ft |
| Maximum:<br>Minimum: | 40<br>--- | 3.5<br>--- | 220<br>--- | 10<br>--- 2/ | ---<br>30 | 300<br>--- 4/ | 5/<br>--- |
| Test condition: | --- | --- | --- | --- | --- | --- | --- |

See footnotes at end of table I.

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GENERAL:

Qualification – Not Required.
This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

AMSC N/A

FSC 5960

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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 1 | | | | | | |
| Total grid current | 1266 | t = 120;<br>Rk = 825 ohms 6/ | Ic1 | 0 | -5.0 | μA dc |
| Electrode current (1) (anode) | 1256 | Rk = 825 ohms 6/ | Ib | 28 | 37 | mA dc |
| Electrode current (2) (anode) | 1256 | Ebb = 4,000 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>Rp = 2 MegΩ 7/ | Ib | 0 | 1.0 | mA dc |
| Electrode current (screen) | 1256 | Rk = 825 ohms | Ic2 | --- | 3.0 | mA dc |
| Conformance inspection, part 2 | | | | | | |
| Pulse operation | --- | Ebb = 4,000 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 155 v 7/ 8/ | --- | --- | --- | --- |
| Grid-pulse operation | 1356 | Ebb = 420 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 155 v 8/<br>(test in accordance with figure 4) | ib | 5.0 | --- | a |
| Peak screen current | --- | Ebb = 420 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 155 v 8/<br>(test in accordance with figure 4) | ic2 | --- | 4.0 | a |
| Peak grid current (1) | --- | Ebb = 420 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 155 v 8/<br>(test in accordance with figure 4) | ic1 | --- | 2.0 | a |
| Peak grid current (2) | --- | Ebb = 420 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 52 v 8/ | ic1 | 30 | --- | ma |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 2 - Continued | | | | | | |
| Grid current operation | --- | Eb = 250 V dc;<br>Ec2 = 150 V dc;<br>Ec1 = -6 V dc through a 0.5 MegΩ resistor;<br>Ef = 6.5 V 9/ | Ic1 | --- | 15 | μA dc |
| Low-frequency vibration | 1031 | Eb = 250 V dc;<br>Ec2 = 100 V dc;<br>Ef = 12.6 V;<br>Ec1 = -10 V dc;<br>Rp = 2,000 ohms | Ep | --- | 500 | mV ac |
| Heater current | 1301 | | If | 1.55 | 1.85 | A |
| Transconductance | 1306 | Rk = 825 ohms | Sm | 5,000 | 7,000 | μmhos |
| Heater-cathode leakage | 1336 | Ehk = 450 V dc | Ihk | --- | 50 | μA dc |
| Direct-interelectrode capacitance | 1331 | No shield | Cgp | --- | 1.2 | pF |
| | | | Cin | 13 | 21 | pF |
| | | | Cout | 7.5 | 12.5 | pF |
| Shock | 1041 | 450 G;<br>Ehk = 450 V dc;<br>Ec1 = -10 V dc;<br>Rp = 2,000 ohms;<br>Eb = 250 V dc;<br>Ec2 = 100 V dc | --- | --- | --- | --- |
| Vibration-fatigue | 1031 | 5 G; F = 50 to 500 Hz;<br>Ef = 6.3 V;<br>no other voltages applied 10/ | --- | --- | --- | --- |
| Shock and vibration-fatigue-test end points | --- | | | | | |
| Heater-cathode leakage | 1336 | Ehk = 450 V dc | Ihk | --- | 100 | μA dc |
| Grid-pulse operation | 1356 | Ebb = 420 V dc;<br>Ecc2 = 800 V dc<br>Ecc1 = -150 V dc;<br>egy = 155 v 8/<br>(test in accordance with figure 4) | ib | 3.75 | --- | a |
| Secureness of base, cap, or insert | 1101 | | --- | --- | --- | --- |
| Base material insulating quality | 1216 | | --- | --- | --- | --- |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 2 - Continued | | | | | | |
| Base pin solder depth | 1111 | | --- | --- | --- | --- |
| Permanence of marking | 1105 | | --- | --- | --- | --- |
| Conformance inspection, part 3 | | | | | | |
| Heater-cycling life | 1506 | Ef = 7.0 V;<br>Ehk = 300 V ac;<br>Ec2 = Eb = 0;<br>1 minute "on"<br>4 minutes "off" | --- | --- | --- | --- |
| Heater-cycling life-test end points | --- | | | | | |
| Heater cathode leakage | 1336 | Ehk = 450 V dc | Ihk | --- | 150 | μA dc |
| Heater current | 1301 | | If | 1.4 | 2.0 | A |
| Life test | --- | Group B;<br>pulse operation;<br>Ebb = 3,500 V dc;<br>Ecc2 = 800 V dc;<br>eo = 300 v;<br>tp = 4 μs;<br>prr = 250;<br>R1 = 500 ohms ± 5 percent<br>11/ | --- | --- | --- | --- |
| Life-test end point (500 hours) | | | | | | |
| Grid-pulse operation | 1356 | Ebb = 420 V dc;<br>Ecc2 = 800 V dc;<br>Ecc1 = -150 V dc;<br>egy = 155 v 8/<br>(test in accordance with figure 4) | ib | 3.75 | --- | a |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

NOTES:

1/ Instantaneous anode voltage due to transient shall not exceed 5,000 volts. Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere. With a screen potential not exceeding 400 V dc and when no instantaneous voltage due to transient is present (essentially resistance anode load), a maximum anode potential of 4,500 V dc may be used.

2/ Instantaneous negative grid voltage due to transient shall not exceed -500 volts. Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere.

3/ Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere.

4/ Cooling shall be employed if the additive effects of simultaneous operation at high altitude and high dissipation or proximity to other heat sources in the equipment tend to cause the maximum bulb temperature to be exceeded.

5/ See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

6/ This test shall be performed at the conclusion of the holding period.

7/ The circuit shown on figure 3, or the equivalent, shall be used for high-voltage anode current (2) test and high voltage pulse operation test.

8/ After a minimum preheat fo 30 seconds using Ef = 6.3 volts only, the tube shall be tested under the following conditions: A substantially rectangular pulse with a width of 4 μs and a prr of 250 shall be applied to the grid of the tube under test. Initial arcing may be tolerated, but the tube shall be free of such arcing during final 30 seconds. of a 1-minute test.

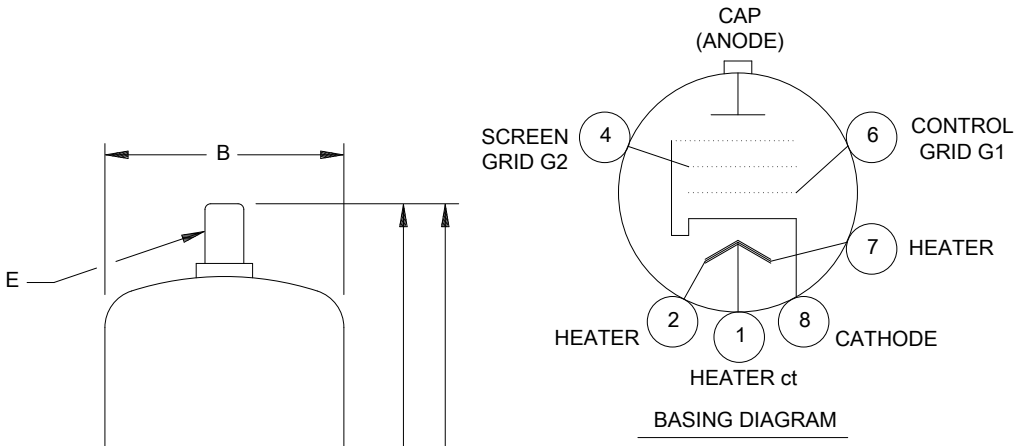
9/ All voltages apply simultaneously. There shall be no preheating. The grid current is read after 5 minutes.

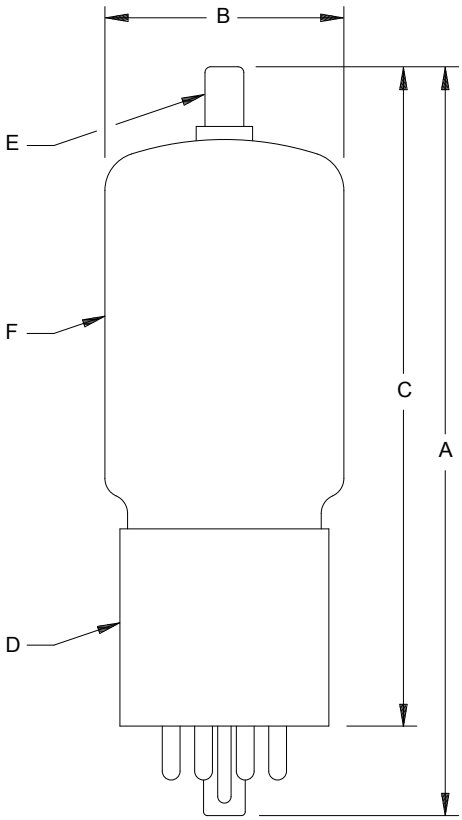
10/ This test shall be performed with an average acceleration level equal to the specified value while being swept through the frequency range of 50 to 500 to 50 Hz. The tubes shall be mounted in each of three planes X, Y, and Z for one-third of the total excitation time. The frequency in Hz shall be scanned continuously up and down through the entire frequency range at the rate of approximately 2 minutes. The total excitation time for the test shall be 96 hours.

11/ The circuit shown on figure 5, or equivalent, shall be used for life-test operation.



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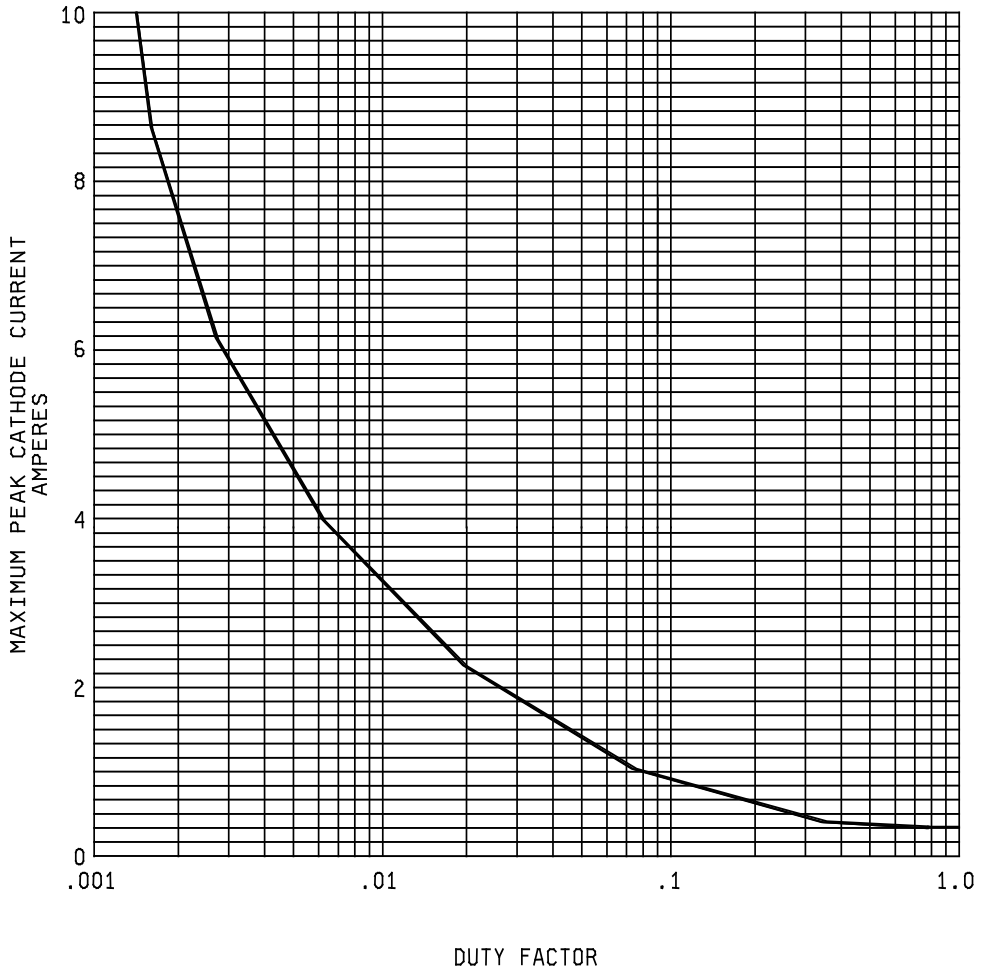




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | --- | 4.688 | --- | 119.08 |
| B | 1.438 | 1.563 | 36.53 | 39.70 |
| C | 3.750 | 4.125 | 95.25 | 104.78 |
| Conformance inspection, part 3 (periodic check) | | | | |
| D | Base: B8-11 (low-loss phenolic) | | | |
| E | Cap: C1-2 | | | |
| F | Envelope: T12 | | | |

FIGURE 1. Outline drawing of electron tube type 3D21WB.



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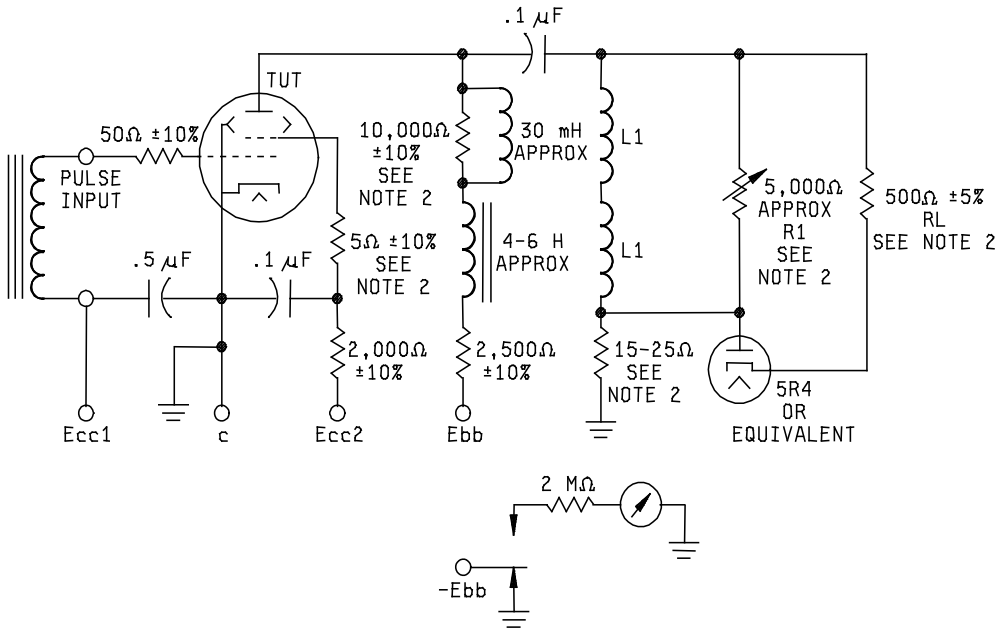



NOTES:
1. Duty factor (10,000 μs averaging time) is defined as the ratio of "on" time in μs to 10,000 μs.
2. "On" time is defined as the sum of the duration of all individual pulses which occur during any 10,000 μs interval.

FIGURE 2. Rating chart.



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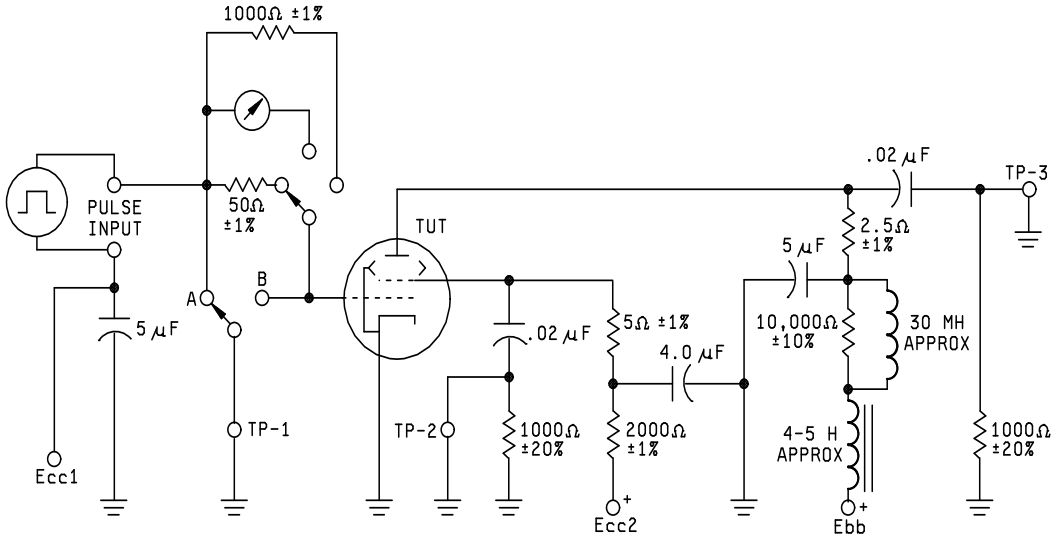

NOTES:

1. $L_1$ = 5.5 mH approximately (Sickles SC101A, or equivalent.)
2. Use noninductive resistor.
3. Use paper or mica dielectric capacitors.
4. Anode current is read on a microammeter in series with the negative lead of the power supply and is shown at point E. The 2-megohm resistor and meter are inserted in the negative lead as shown for the anode current test only. $L_1$ and $R_1$ adjusted for the voltage backswing = 50 percent of forward swing.

FIGURE 3. Test circuit.



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NOTES:

1. Peak anode current (ib) is read on a calibrated oscilloscope across a 2.5-ohm sensing resistor (TP-3).
2. Peak screen current ($ic_2$) is read on a calibrated oscilloscope across a 5.0-ohm sensing resistor (TP-2).
3. Peak grid current (1) ($ic_1$) is read on either a calibrated oscilloscope across a 50-ohm sensing resistor (TP-1) or a milliammeter calibrated for peak reading with duty cycle of 0.001).
4. Peak grid current (2) ($ic_1$) is read on either a calibrated oscilloscope across a 1,000-ohm sensing resistor (TP-1) or a milliammeter calibrated for peak reading with duty cycle of 0.001).
5. egy = e in switch 'B' position.

$$ic_1(1) = \frac{e(switch\ position\ A) - e(switch\ position\ B)}{50}$$

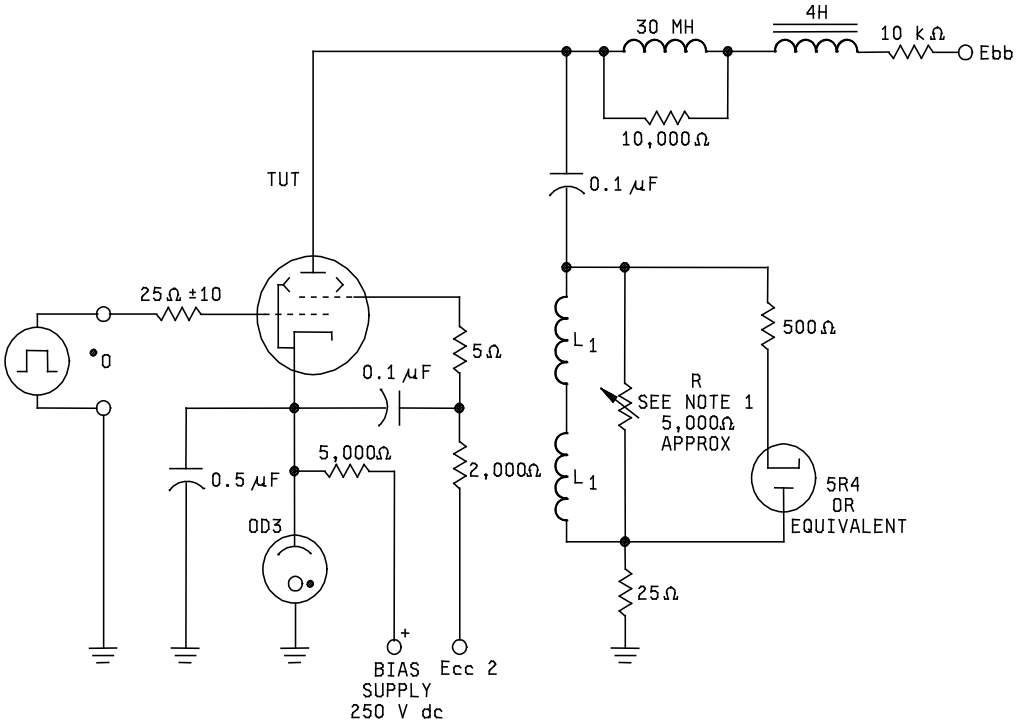
$$ic_1(2) = \frac{e(switch\ position\ A) - e(switch\ position\ B)}{1{,}000}$$

6. Use noninductive resistors.
7. All capacitors shall be mica or paper dielectric with suitable voltage ratings. TP-1, TP-2, TP-3 to suitable wide-band oscilloscope with low-capacity probe.

FIGURE 4. Test circuit.



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NOTES:
1. Adjust for 50 percent backswing.
2. L1 = 5.5 mH approximate (Sickles SC101A, or equivalent).

FIGURE 5. Test circuit.



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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

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